

MAT 426, Class 18: Orthogonal Diagonalization & Quadratic Forms

THEOREM *Let A be an $n \times n$ real matrix. Then A is symmetric if and only if it is orthogonally diagonalizable.*

Proof.

Spectral Decomposition of A

Example 1. Find the spectral decomposition of $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$.

Solution. Last class we showed that $A = QDQ^T$ where

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{bmatrix} \quad D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Definition. A **quadratic form** on $\mathbb{R}^n$ is a function $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ where A is an $n \times n$ _____ matrix.

Example 2. Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. For each of the following matrices, compute $\mathbf{x}^T A \mathbf{x}$.

(a) $A = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$

(b) $A = \begin{bmatrix} 2 & -3 \\ -3 & 6 \end{bmatrix}$

Example 3. Write the given quadratic form Q as $\mathbf{x}^T A \mathbf{x}$.

(a) $Q(\mathbf{x}) = x_1^2 - 8x_1x_2 - 5x_2^2$

(b) $Q(\mathbf{x}) = 5x_1^2 + 3x_2^2 + 2x_3^2 - 2x_1x_2 + 6x_2x_3$

PRINCIPAL AXES THEOREM Let A be an $n \times n$ _____ matrix. Then, there exists an _____ change of variable, $\mathbf{x} = P\mathbf{y}$ that transforms the quadratic form $\mathbf{x}^T A \mathbf{x}$ into a quadratic form $\mathbf{y}^T D \mathbf{y}$ with _____ term.

Proof.

Example 4. Make a change of variables that transforms the quadratic form $Q(\mathbf{x}) = x_1^2 - 8x_1x_2 - 5x_2^2$ into a quadratic form with no cross-product term.

Definition. A quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ is classified as one of the following:

- (1) **positive definite** if _____ for all $\mathbf{x} \neq \mathbf{0}$.
- (2) **positive semidefinite** if _____ for all $\mathbf{x}$.
- (3) **negative definite** if _____ for all $\mathbf{x} \neq \mathbf{0}$
- (4) **negative semidefinite** if _____ for all $\mathbf{x}$
- (5) **indefinite** if $Q(\mathbf{x})$ takes on both _____ and _____ values

A symmetric matrix A has the same classification as the associated quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$.

THEOREM Let A be an $n \times n$ symmetric matrix. The quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ is

- (a) positive definite if and only if all of the _____ of A are _____
- (b) positive semidefinite if and only if all of the eigenvalues of A are _____
- (c) negative definite if and only if all of the eigenvalues of A are _____
- (d) negative semidefinite if and only if all of the eigenvalues of A are _____
- (e) indefinite if and only if A has both _____ and _____ eigenvalues